

Semantic Epistemic Risk versus Confidence-Based Active Retrieval During Generation

Working Draft

August 2026

Abstract

Active retrieval during generation is established prior art: FLARE predicts upcoming content and retrieves when that prediction contains low-confidence tokens [1]. We ask a narrower question: is model uncertainty equivalent to epistemic risk? Using one controlled evidence source, we compare a documented FLARE-like confidence controller with transparent semantic rules that detect stale, source-dependent, or otherwise unsupported imminent commitments. The experiment measures confidence on the actual checkpoint and deliberately populates all four confidence-risk regimes. Any contribution is evidence that semantic risk catches needs missed by uncertainty at competitive cost, not the idea of generation-time retrieval itself.

1 Research Question

When confidence and epistemic risk are measured separately, does semantic risk-triggered retrieval improve the unsupported-claim-cost frontier over FLARE-like confidence triggering, model-only generation, upfront RAG, and explicit search/tool calling?

2 Mechanism

$$Q \rightarrow y_{1:t} \rightarrow \hat{y}_{t+1:t+k} \rightarrow \{T_{\text{conf}}, T_{\text{sem}}\} \rightarrow \text{retrieve} \rightarrow \text{evidence} \rightarrow y_{t+1:}$$

The model need not emit a search API call. The FLARE-like condition predicts or inspects upcoming generation, computes token/model uncertainty, retrieves with the upcoming content below a fixed confidence threshold, then regenerates or continues with evidence. We report every simplification relative to canonical FLARE. The semantic condition uses only transparent lexical, syntactic, entity, temporal, freshness, and source-dependence rules.

3 Single Controlled Source

Use a versioned fact store or relational database as the primary source. It must contain stable facts, changed/recent facts, unavailable facts, contradictory records, plausible distractors, and facts whose relevance becomes apparent only after intermediate reasoning. Web retrieval may be used only as a secondary ecological demonstration.

4 Trigger Design

Paper 1.5 permits only transparent, non-learned triggers:

- lexical cues such as “latest”, “current”, “in 2026”, “highest”, “most sales”;
- entity/attribute patterns;
- temporal expressions;
- narrow sentence-prefix templates indicating an imminent factual value.

5 Measured Confidence–Risk Design

Neural uncertainty is not an evidence-need label. Examples are assigned only after measuring confidence on the evaluated checkpoint:

Confidence	Epistemic risk	Expected diagnostic behavior
High	Low	continue without retrieval
Low	Low	expose confidence-driven over-retrieval
High	High	test confident-hallucination catch rate
Low	High	both controllers should retrieve

Thresholds are fit on development data and swept at evaluation; quadrant labels must not be assigned from researcher intuition.

6 Prospective and Retrospective Modes

Prospective retrieval. Interrupt before a factual value is emitted.

Retrospective verification. Allow a candidate claim, then retrieve before treating the claim as supported.

7 Epistemic State

Return compact typed status alongside evidence:

- supported;
- contradicted;
- unverified;
- stale;
- ambiguous;
- source conflict.

All records retain source identity and timestamps where available.

8 Tasks

Include confidently wrong or stale facts, low-confidence completions needing no evidence, stable high-confidence facts, and low-confidence externally grounded facts. Also include multi-step answers where a need emerges late, freshness-sensitive questions, unavailable evidence requiring abstention, and hard distractors where neither trigger should fire.

9 Baselines

1. LLM only.
2. Matched anti-hallucination prompt.
3. Upfront RAG with identical source access.
4. Explicit retrieval/tool calling.
5. FLARE-like confidence trigger.
6. Semantic heuristic trigger.
7. Confidence OR semantic trigger.
8. Confidence AND semantic trigger where meaningful.
9. Retrospective verification.
10. Oracle trigger and query.

10 Metrics

Report final accuracy, unsupported-claim rate, trigger and retrieval precision and recall, confident-hallucination catch rate, false retrieval for low-confidence cases without evidence deficits, retrieval rate in each measured confidence–risk quadrant, uncertainty around thresholds, correction and abstention quality, evidence-status calibration, tokens, calls, and latency. Plot hallucination reduction against retrieval cost with threshold sweeps and Pareto frontiers.

11 Falsification

Semantic triggering fails to add value if the FLARE-like condition catches the same cases at equal or lower cost, the semantic rules duplicate uncertainty, high-confidence/high-risk cases are immaterial, false triggers erase reliability gains, or the rules effectively encode answers.

12 Scope

One retrieval source only. No learned trigger, no multi-source router, no symbolic reasoning, no backtracking, no PRA/native KV dependency.

References

- [1] Z. Jiang et al. Active retrieval augmented generation. *EMNLP*, 2023. <https://arxiv.org/abs/2305.06983>.